Description

**** I reserve the right to modify the course content, schedule, topics, policies, etc. outlined in this syllabus.****

Basic Information

Rose-Hulman Course Catalog Description

An introduction to mathematical statistics. Review of distributions of functions of random variables. Moment generating functions. Limiting distributions. Point estimation and sufficient statistics. Fisher information and Rao-Cramer inequality. Theory of statistical tests.

Topics We Will Cover

Methods to obtain statistical estimators; Statistical inference methods; Inference in linear regression; Exponential family of distributions

Prerequisite

{MA382 (Introduction to Statistics)} OR {MA381 (Introduction to Probability) and MA223 (Engineering Statistics)}

Role of the Prerequisite

The prerequisite course implies that students have a firm grasp on basic statistical methods and exposure to essential probability theory before entering MA481. In addition, mastery of integrals and derivatives will be expected. Other courses which may be useful (but not required) include linear algebra, real analysis (ability to construct well-written proofs), and statistical computing.

No textbook required

There isn't an existing textbook that I feel is worth the cost while also containing enough topics relevant to our course to require a text. However, it is strongly recommended that you consider using a textbook as reference when working through course material.

Below are recommended texts for this course. I will provide relevant textbook sections alongside our course notes.

Rice. *Mathematical Statistics and Data Analysis*. (2007). 3rd edition. Online pdf available for free.

Keener. *Theoretical Statistics*. (2010). 1st edition. Online pdf available through the RHIT library

Statistical Software

We will exclusively use R for data analysis in this course. Please see the Getting Started in MA481 section.

Hybrid Format

Face-to-Face Class Meetings

Monday and Thursday @ 9am in G220

The face-to-face class meetings will have a similar feel to a traditional lecture (e.g. instructor-led content and activities). The in-person meetings are required and all students need to plan on attending these. However, I encourage students who are ill to stay home and not come to class. If you are sick, please let me know as soon as possible. Unfortunately I do not have videos for every single face-to-face class day. With enough advance notice, there are some days where I might be able to record class and provide video access afterwards.

Some of our face-to-face days will be associated with completing examples. On those days, there won't be lecture; you will work together as a class to formulate solutions to questions.

Online Days

Tuesday and Friday, asynchronous

The online days will mostly be associated with using simulations to connect the theoretical ideas to what occurs in practice. Students may need to fill in notes packets and follow along to produce a set of results using simulations in R.

Important: Although we will not meet at a specific time, it is expected that students work through the provided material at some point during the specified day. The face-to-face meetings will operate under the assumption that the previous days' online material was completed.

All course notes packets will contain a set of objectives that students should be able to complete by the end of the session. These objectives are tied to overall course objectives. The overall course objectives are then assessed through various assignments.

Office Hours

Office hours in DL110-B: 10:00 - 10:50 AND 12:00 - 12:50 on Monday & Thursday

- My office is inside the Engineering Management/Graduate Studies/Lifelong Learning office suite. I'm first door in the hallway on the left. If you've never been there, it is in Moench hall to the right of the "commons" entrance.

Availability other times:

- I try to check email regularly during work hours: 8am-8pm Monday through Friday. *My availability outside of typical hours may be limited*. Although I do use Microsoft Teams, I do not check it as often.
- I attempt to keep my Outlook calendar updated. If you cannot come to in-person office hours or have questions on other days of the week, please do not hesitate to reach out to me over email. We can set up an ad-hoc meeting either in person or on Teams.
- I will make every effort to try to meet with you upon request. I do request your patience and flexibility, as there may be times when I cannot meet with you immediately.

Assessment Structure

This course was created using "backwards design." This implies that lectures, activities, and assessments are created based on course goals and objectives.

Assessments are mapped to overall course goals and objectives.
This course has two overarching goals. Each type of assessment has one or more associated goals. (For example, Simulations are associated with Goal #1.) Thus, the importance or use of any graded assignment in this class should be clear, since the assignments directly assess valuable objectives.

Each module has 3 assignments.
Students can work together but submit their own solutions to complete the Simulations and Proofs assignments. After both module assignments pass, you unlock your Module Exam. The Module Exam is completed individually.

Students aiming for a B or A in the course will complete a capstone project.
The capstone project asks students to learn about a topic outside of our course materials in-depth. Students will complete a set of simulations and/or a report as their project deliverable.

Simulations

Simulations assignments are written to align with the following course goal and objectives.

Goal #1: Students will understand how simulations are related to theoretical statistical results.

In order to achieve this goal, you will...

1.a Write code to implement a statistical simulation.
1.b Identify aspects within simulation code that are associated with theoretical assumptions.
1.c Visualize the results of a simulation using informative graphics.
1.d Summarize simulation results to support a given theorem.
1.e Update simulation code to explore results when assumptions of a theorem are not met.

What to expect with simulations

There will be 10 simulations assignments, occurring weekly.

- On Tuesday of each Module, you will learn about a simulation topic and follow along with an instructor-led example.
- The weekly simulation assignment will be an extension of the content presented on Tuesday that is relevant to the module topic.
- You should be able to start working on the simulation assignment after completing the second day of the module (Tuesday)
- Simulations assignments will be collected via Moodle.

In order to pass the Simulations assignment, you need to meet all listed rubric criteria.

- There is no time limit on the assignment, but you should plan to submit the assignment by the end of each module (typically, 11:59pm Eastern time, Sunday after the module completes).
- You may use any course materials as reference when working on the assignment.
- The assignment will be available for you to view and work on throughout the entire module week.

Each student should complete their own assignment.

- It is permissible to work with other students. However, any of the following may be considered academic misconduct:
    - utilizing or copying another students' code
    - utilizing or copying another students' discussion
    - utilizing or copying another students' collected data

- Students engaging in academic misconduct will receive a 0 for the entire assignment for the offense and will not be allowed to resubmit the assignment. Academic Affairs will also be notified.

Students may resubmit any Simulations assignments that are unsuccessful on the first attempt. Please see the "Resubmissions" section for more information.

Proofs

Proofs assignments are written to align with the following goal and objectives.

Goal #2: Students will know how to write statistical proofs.

In order to achieve this goal, you will...

2.a Write a proof to establish a property of a statistical estimator.
2.b Compare and contrast modes of convergence.
2.c Associate a statistical method with various statistical properties such as consistency, unbiasedness, and asymptotic normality.
2.d Derive the formulas for different types of statistical estimators.
2.e Justify steps within a provided proof.
2.f Identify incorrect statements within a provided proof and explain why the statements are incorrect.

What to expect with Analyses

There will be 10 Proofs assignments, occurring weekly.

- Each assignment will contain two questions or proofs to be completed.
- Proofs assignments will be collected using Moodle.
- The material on the assignment is from the associated module. You should be able to complete the assignment once the module material is finished.

In order to pass the Proofs, you need to meet all listed rubric criteria.

- There is no time limit on the assignment, but you should plan to submit the assignment by the end of each module (typically, 11:59pm Eastern time, Sunday after the module completes).
- You may use any course materials as reference when working on the assignment.
- The assignment will be available for you to view and work on throughout the entire module week.

Each student should complete their own assignment.

- It is permissible to work with other students. However, any of the following may be considered academic misconduct:
  - utilizing or copying another students' code
  - utilizing or copying another students' discussion
  - utilizing or copying another students' collected data
- Students engaging in academic misconduct will receive a 0 for the entire assignment for the offense and will not be allowed to resubmit the assignment. Academic Affairs will also be notified.

Students may resubmit any Proofs assignments that are unsuccessful on the first attempt. Please see the "Resubmissions" section for more information.

Module Exam

Module Exam assignments are written to assess the same goals and objectives as the Simulations and Proofs assignments.

What to expect for Module Exams

There will be 10 Module Exam assignments, occurring weekly.

- Each module exam will contain one simulation question and two proofs questions.
- Module exams will be collected using Moodle.
- The material on the Module Exam is from the associated module. If you have completed all material for the module, you should be able to complete the Module Exam.

Module Exams only become available after both the Simulations and Proofs assignments for the module pass.

- This is similar to "unlocking" the exam by demonstrating basic knowledge of the module before attempting the exam.
- There is no time limit on the assignment. It is strongly recommended that you attempt to submit it by 11:59pm Eastern time, Wednesday after the module has completed so your coursework does not get too far behind.
- You may use any course materials as reference when working on the exam.

Module Exams are an individual effort. That is, students do not work together to complete these assignments.

- Students engaging with academic misconduct will receive a 0 for the entire assignment for the offense and will not be allowed to resubmit the assignment. Academic Affairs will also be notified.

Students may resubmit Module Exams up to 3 times after the initial attempt. Please see the "Resubmissions" section for more information.

Capstone Project

Those students aiming for an A or B in the course must successfully complete a capstone project. The project asks you to read a peer-reviewed journal article, textbook section, or a post on a well-established website (provided by the instructor) regarding a topic of your choosing (from a list approved by the instructor).

You will then complete a report and/or set of simulations related to your topic.

Please see the MA481 Capstone Project section in our course site for more information.

Grades

I want this course to create a strong foundation in statistical experimental design for you. **In order to help you achieve this, I will be implementing a version of "specifications-based grading."** That is, instead of taking a weighted average of points earned on a series of assignments throughout the term, you will earn grades based on the assignments that you successfully complete.

Successfully completed assignments earn credit toward your overall course grade. Each assignment is graded as pass (1) / fail (0). In order to help you achieve a passing score, very clear expectations will be provided on every assignment, often via a rubric.

You will also notice that I am not expecting perfection. For example, a student may achieve an A in the course without having to pass every single course assignment. However, successfully completed/passing assignments will be based on clear, high standards. You will also have the chance to resubmit some assignments that did not receive credit on the first attempt. This awards you for going back to learn the material, even if it did not "click" the first time through.

In the end, you have complete control over your grade in the course; I will provide you with the necessary tools and feedback to help you achieve your desired grade.

Letter Criteria

Final letter grades in this course will be assigned based on the mastery of course material demonstrated throughout the term.

- **There are a total 10 Module Exams during the term** (one per module).
- **There are 10 Simulations and 10 Proofs assignments** (one of each per module). Successful completion of both assignments unlocks the module exam.
- **There is one Capstone Project** (occurring during week 10).

In order to earn a grade of "D"

You must meet **all** of the criteria listed below:

- Successfully complete 5 Module Exams
- Successfully complete one additional Simulations assignment (beyond the 5 already passed for your module exams)
- Successfully complete one additional Proofs assignment (beyond the 5 already passed for your module exams)

In order to earn a grade of "D+"

You must meet **both** of the criteria listed below:

- Complete all criteria associated with D
- Pass at least one additional course assignment (Proofs or Simulations)

In order to earn a grade of "C"

You must meet **all** of the criteria listed below:

- Successfully complete 6 module exams
- Successfully complete one additional Proofs assignment (beyond the 6 already passed for your module exams)
- Successfully complete one additional Simulations assignment (beyond the 6 already passed for your module exams)

In order to earn a grade of "C+"

You must meet **both** of the criteria below:

- Complete all criteria associated with C
- Pass at least one additional course assignment (Proofs or Simulations)

In order to earn a grade of "B"

You must meet **all** of the criteria listed below:

- Successfully complete 7 module exams
- Successfully complete one additional Proofs assignment (beyond the 7 already passed for your module exams)
- Successfully complete one additional Simulations assignment (beyond the 7 already passed for your module exams)
- Successfully complete the Capstone Project, Tier 2.

In order to earn a grade of "B+"

You must meet **both** of the criteria below:

- Complete criteria associated with B
- {Pass at least one additional course assignment (Proofs or Simulations)} **OR** *{Successfully complete the Capstone Project Tier 1}*

In order to earn a grade of "A"

You must meet **all** of the criteria listed below:

- Successfully complete 8 module exams
- Successfully complete one additional Proofs assignment (beyond the 8 already passed for your module exams)
- Successfully complete one additional Simulations assignment (beyond the 8 already passed for your module exams)
- Successfully complete the Capstone Project, Tier 1.

An "F" is assigned when the criteria to earn a "D" are not achieved.

Resubmissions

Life happens. We all have different priorities, and at times the priorities may not easily align with the timing for class assignments. I recognize that this happens with the busy schedules that students have at RHIT and want to give every student adequate opportunity to succeed. Moreover, I value the learning experience of re-examining material that was not mastered on the first encounter.

All students will have the opportunity to resubmit most course assignments that were not initially successful. There are many opportunities to resubmit throughout the term.

Here's how resubmissions work

- Each Monday, Thursday, and Friday, Dr. Heyman will grade any assignments that are submitted in Moodle by 8am that day.
    - Dr. Heyman can automatically see when assignments are updated in Moodle. Thus, she does not need email notification of a resubmission. However, if your assignment is resubmitted and you do not see any updated feedback or grade in a timely manner, it is completely fine to check in with Dr. Heyman about this.

- **Analysis assignments that are not successful on the first attempt can be resubmitted any time during the term.**
    - There are 9 analysis assignments possible during the term. You can resubmit these at any time after your initial submission.
    - If you miss the initial submission deadline, you can submit a late attempt as a "resubmit"
    - You can resubmit the analysis assignments as many times as you need until the solution is successful.

- **Module Exam assignments that are not successful on the first attempt can be resubmitted up to 3 times during the term.**
    - Module Exam assignments are completed as an individual. Students that work together on exams (whether it is the first attempt or any resubmission) will not receive credit for the exam and also be reported under the Mathematics department academic integrity policy.
    - There are 9 module exams possible during the term. You can resubmit these at any time after your initial submission.
    - Module exams become available to you after passing both the Quiz and Analysis from the Module.

    - You can resubmit any Module Exam up to 3 times (thus, any Module Exam would have a maximum of 4 possible attempts)

- **Class Experiment Reports that are not successful on the first attempt can be resubmitted once.**
    - There are a total of 4 experiment reports during the term.
    - You are only allowed a single resubmission per report. If the report is submitted after the deadline, the late submission counts at the resubmission for that report.

- **There are no resubmissions associated with Quizzes and the Final Project.**

Participation

Rose-Hulman Institute of Technology is **committed to being an inclusive community** in which the multiplicity of values, beliefs, intellectual viewpoints, and cultural perspectives enrich learning and inform scholarship. I want our classroom to be an open and safe environment to share and discuss ideas. Thus, it is expected that students and the professor are good citizens during and outside of class, treating each other with mutual respect.

- **Discussion and questions are encouraged during class**. Feel free to speak up at any time during lecture sessions. My classroom is a fairly informal atmosphere.
- If anything occurs during or outside of our class (for example, a microaggression) which makes you feel uncomfortable, please reach out to me. I want all students to feel comfortable in my classroom, and it is my intention to foster both equality and equity.
- If there are any instances where a student continues disrespectful behavior after discussion with me, to the point that the behavior is disrupting our classroom experience, a penalty may be incurred.
- Lecture attendance is integral to success in this course. It is easy to fall behind from missing 1 or 2 class sessions.
    - With the hybrid format, it is especially important to keep on schedule, since we have fewer face-to-face meetings.
- Please do not use your cell phone during class. It is disruptive to other students and the professor. Laptop use should be responsible and appropriate to our class activities.

Institute Policies

Students with Accessibility Needs

Rose-Hulman Institute of Technology strives to make all learning experiences accessible to students. If you anticipate or experience academic barriers based on accessibility issues, please feel free to communicate those needs and register with Student Accessibility Services.

Student Accessibility Services will work with you understand the process and to determine what accommodations are most appropriate for your individual situation. Visit the Accessibility Services website for more information. Please note that accommodations are not retroactive and accommodations cannot be provided until verified.

Please contact Student Accessibility Services for more information at HMU 156, 812-877-8040, or eaton1@rose-hulman.edu.

Please note that it is the student's responsibility to request any approved, documented academic accommodations (such as extra time) at least three days in advance of exams.

Academic Integrity

Plagiarism, offering, and accepting solutions to assessments when a student is required to work individually are cases of academic misconduct. These instances are taken seriously by the professor and university, and will be reported. Anyone found cheating will not be permitted to withdraw and will be (appropriately) heavily penalized in the course. Academic affairs will be informed.

The Student Handbook and Rose-Hulman's Academic Rules and Procedures describe penalties and processes invoked as a consequence if academic misconduct (such as cheating, plagiarizing, or interfering with the academic progress of other students) takes place. It is the responsibility of each student to know and follow Rose-Hulman's rules about academic integrity.

Incompletes

Incompletes are granted only when a student is forced to miss several days of class due to extraordinary circumstances such as a documented confining illness or family emergency. To receive an incomplete grade, students should obtain approval from the professor before the last day of class.

No incompletes will be given unless a prior written agreement with the professor exists, following typical Rose-Hulman policy. In this course, an "I" grade will be given only in cases of extreme hardship. Poor performance in the course is not grounds for an incomplete. An incomplete shall be recorded as an "F" if the work is not completed within the time agreed upon by the professor and student.

Emergency Information

To receive email or text messages regarding emergency situations that may impact campus and, possibly, the delivery of classes, register for RAVE alerts *and/or follow* @Rose-HulmanAlert on Twitter. *Any announcements about the Institute's ability to offer classes will be shared on* Rose-Hulman's website.